uCNC RP2350B Motherboard

Generated board documents:

- Combined board review
- Front artwork and assembly layer
- Mirrored rear artwork

Board specification

Item	Specification
Product	LeanCam NCC / uCNC RP2350B Motherboard
Status	Routed pre-release prototype; final cleanup in progress
Controller	Socketed WeAct Studio RP2350B Core Board
PCB outline	200 mm x 150 mm
PCB construction	Two copper layers, 1.6 mm finished thickness
Mounting	Four M3 clearance holes
Assembly	Main assembly on the front; J22, C57 and JP2-JP4 on the back
Input supply	24 V DC nominal machine supply, documented operating range 12-24 V
Main 5 V rail	LMR36510 synchronous buck, 1 A
Backlight rail	Second LMR36510 buck, 12 V, 1 A
Logic supply	AMS1117-3.3, +3V3
Wi-Fi supply	Separate AMS1117-3.3, +3V3_WIFI
Analog supply	L78L12, +12V_AO
Isolated field supply	Traco TBA2-0512, +12 V isolated wetting rail
Board ID	MOD1, the socketed RP2350B module

The board is a machine-control carrier, not a safety-rated controller. Hazardous output power must be removed through an independent hard-wired safety circuit.

Digital I/O architecture

Function	Capacity	Implementation
Isolated PLC inputs	16	Four LTV-847S quad optocouplers with four 4x4.7 k pull-up networks
Dedicated safety/control inputs	3	E-stop, HOLD and START through individual EL3H7 DIP-4 optocouplers
3.3 V expansion inputs	8	Two four-channel headers, J9 and J27
Keyboard inputs	8	KIN0-KIN7 through the input shift chain
General field outputs	16	DO0-DO15 through ULN2003A outputs and two BSS138 low-side stages
Motion outputs	15	Five EN/STEP/DIR axes plus one AUX output
Keyboard outputs	8	KOUT0-KOUT7 through the output shift chain
Encoder inputs	3	A/B/Z channels with 5 V to 3.3 V conditioning
Analog inputs	3	0-10 V, resistor-divided into RP2350 ADC inputs
Analog outputs	4	0-10 V through TLV2374 and RC filtering

The digital chains are 32 bits wide:

- Input: four cascaded 74HC165 registers, eight bits each.
- Output: four cascaded 74HC595 registers, eight bits each.
- Shift data uses GPIO20, GPIO21 and GPIO22; the input return is GPIO23.
- E-stop, HOLD and START connect directly to GPIO37, GPIO38 and GPIO39.

The complete GPIO and shift-bit mappings are in `documentation/gpio-map.md` and `documentation/io-bit-map.md`. Firmware must preserve the documented physical pin permutations.

Isolated field inputs

The 16 PLC inputs support two modes:

- External 24 V wetting.
- Dry contacts powered from the onboard isolated 12 V supply.

JP2, JP3 and JP4 select the wetting source per field bank and safety channel. The isolated field return is `FIELD_COM0` or `FIELD_COM1`; it is not connected to motherboard GND. D56 prevents external field voltage from backfeeding U40.

Each PLC channel includes a 2.2 k series resistor, indicator LED, reverse-protection diode and one LTV-847S optocoupler channel plus one element from a 4x4.7 k logic-side pull-up network. The E-stop, HOLD and START pull-ups remain discrete. Firmware must debounce the sampled contacts.

Connectors

Ref	Function
J1	24 V DC power input
J2	Backlight/CCFL: +12 V, GND, PWM, switched low-side return
J4	SWD programming/debug
J5	Dedicated E-stop loop
J6, J26	HOLD and START buttons
J7, J8, J34, J35	16 isolated PLC inputs
J9, J27	Expansion inputs 0-7, 3.3 V logic only
J10, J11, J28, J29, J30	Axis EN/STEP/DIR outputs
J12, J32	General field outputs DO0-DO15
J33	General-output field supply and GND
J13-J15	Encoders 0-2
J16	Keyboard KIN/KOUT
J17	Keyboard power/reference
J18	I2C0
J19	Full-size SD card, SPI mode
J20	Optional ESP-01/ESP-01S
J22	2x10, 2.00 mm LCD LVDS
J24	Three 0-10 V analog inputs
J25	Four 0-10 V analog outputs
J31	AUX low-side output

Detailed connector pinouts are in `documentation/connectors.md`.

Power and protection

- Main input path: F1 2 A fuse, D1 24 V TVS and two 1 A SS16 series diodes.
- D2 provides reverse-polarity protection for the logic-side input; D3 does the same for the backlight buck.
- U1 converts the protected machine input to the 1 A +5 V rail.
- U3 converts its separately diode-fed input to `+12V_BL` at up to 1 A.
- The WeAct module receives +5 V and uses its own regulated 3.3 V rail.
- U2 independently supplies board logic, SD and LCD logic at +3V3.
- U34 independently supplies the optional ESP module at +3V3_WIFI.
- U37 supplies only the analog-output amplifier, avoiding a local hot spot.
- U40 provides an isolated +12 V wetting rail for dry-contact input mode.

Do not use the analog or isolated field rails to power motors, lamps or high-current solenoids. General field outputs are limited by their external supply and the ULN2003A/MOSFET ratings.

Display and storage

- J22 carries three LVDS data lanes and one clock lane for a 2x10 panel interface. Rows 9-10 and pins 17-20 are not connected.
- U41, a DS90LV047A, is the active LVDS driver population.
- RN26 and RN27 are a mutually exclusive passive LVDS alternative. Do not fit U41 and both resistor networks together.
- There is no HDMI connector or HDMI-specific support circuitry.
- The LCD connector has no backlight contact. J2 carries `+12V_BL`, a 3.3 V `BL_PWM` signal from GPIO35, ground and an optional Q8 low-side return. R131 series-protects only the connector PWM path; R129 branches from the same GPIO-side node to Q8.
- High-current backlight switching and the CCFL inverter remain external.
- The full-size SD card uses 3.3 V SPI and includes card-detect.

PCB and manufacturing data

Metric	Value
PCB footprints	333
BOM component quantity	327
Through-hole / SMD pads	269 / 1018
Through vias	214
Minimum track width	0.12 mm
Minimum drill	0.30 mm

The BOM is in `manufacturing/bom.csv`. This specification is informational and does not replace the KiCad schematic, PCB, DRC or ERC reports.

Verification status

The version documented here was checked with KiCad 10.0.1:

- DRC: no shorts; 4 signal connections remain open after the latest manual placement pass.
- ERC: zero electrical-rule errors.
- Remaining release work: finish manual placement and routing, then clear the 109 standard clearance findings and the remaining silkscreen, dangling copper, thermal and courtyard findings.
- ERC retains 33 library-symbol cache mismatch warnings.

These remaining findings must be reviewed before fabrication. The board is not declared production-ready or safety-certified.